

Project Development Phase

Project Executable Files

Date	02 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

This section documents all executable and deployable files for OptiCrop, so the evaluator can run or deploy the solution using the files and instructions below.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt (Python dependencies)	Yes
4	Database schema / seed files	NA — no database used
5	Environment configuration file (.env.example)	NA — no secrets/API keys used
6	Deployed application URL (if hosted)	https://opticrop.streamlit.app
7	APK / Executable binary	NA — web app, not a mobile/desktop app
8	Dockerfile / Containerization config	NA — not containerized
9	Test files and test results	See Functional and Performance Testing document
10	Demo video or walkthrough	https://drive.google.com/drive/folders/opticrop-demo-video

Step 2: File / Folder Structure

5. Project Development Phase/

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├─ app.py           – Streamlit web application (run this to launch
the site)
├─ model.pkl        – Pre-trained Random Forest model
├─ train_model.py   – Script used to train model.pkl from the
dataset
├─ Crop_recommendation.csv – Training dataset (2200 records, 22 crop
classes)
├─ requirements.txt  – Python dependencies
├─ .gitignore
└─ README.md
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opticrop.streamlit.app
Login Credentials (Demo)	NA — no login required, app is publicly accessible
Platform / Hosting Provider	Streamlit Community Cloud
Repository Link	https://github.com/vijayapadmapriya/OptiCrop
Demo Video Link	https://drive.google.com/drive/folders/opticrop-demo-video

Step 4: Run Instructions

Pre-requisites: Python 3.8+ and pip installed.

Step 1: Clone the repository – `git clone <your-repository-url>`

Step 2: Navigate into the folder – `cd "5. Project Development Phase"`

Step 3: Install dependencies – `pip install -r requirements.txt`

Step 4: Run the app – `streamlit run app.py`

Step 5: Open the browser at the local URL shown in the terminal
(usually `http://localhost:8501`)

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	No authentication / user accounts	By design — the tool is a public single-page utility, no login needed